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# Genetic Algorithm for Vehicle Routing Problem with Time Windows and a Limited Number of Vehicles

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Abstract: This paper improves the mathematical model for the vehicle routing problem with time windows where a limited number of vehicles is given, which not only reflects the feature of the limited number of vehicles in the depot, but also is compatible with VRPTW; redefines the distance between the customers used in the customer clustering, designs the customer clustering assignment algorithm, which can help achieve the initialization of the genetic algorithm and provides a new method for VRPTW customer clustering assignment; proposes an improved chromosome representation on the basis of customer, which can denote the different vehicle number sent by the depot and the customers unserviced; designs the genetic algorithm to solve $m$-VRPTW, and proves the validity of the algorithm by the experiments.

Keywords: clustering analysis, genetic algorithm, limited number of vehicles, vehicle routing problem with time windows

## 1 Introduction

In recent years, with the maturity of VRPTW theory and the rapid improvement of the computer hardware performance, the study of VRPTW covers many fields, and it is widely used in the shipping, aviation, telecommunications, electric power, industrial management, and so on. The heuristic algorithm and intelligent heuristics, such as genetic algorithm[1], tabu search algorithm[2,3] and simulated annealing algorithm [4,5], are proposed, and some good results are achieved.

The above researches about VRPTW, firstly assume that there are enough vehicles to serve all the customers. Actually, the number of vehicles is limited in the logistics company. In practice, there may be some days that the customers' demands suddenly increase, but due to the limited number of vehicles, there are no enough vehicles to service all the customers, which requires that the primary objective of the vehicle routing is no longer the shortest distance or minimum cost, but the greatest number of the customers serviced to pursuit the highest degree of customer satisfaction. For the VRPTW with a limited number of vehicles, the related researches are only found in Lau H. C.[6] and Deng Wei[7]. Lau H. C.[6] definites the $m$-VRPTW problem, and solves it using nested tabu search algorithm for a two-stage solution. Deng Wei[7] establishes the $m$-VRPTW mathematical model, whose first goal is to serve the customers at the greatest number and second goal is to get the total distance at the least cost, and proposes the route construction heuristic algorithm and the Tabu Search algorithm for the problem.

At present, the literature that $m$-VRPTW is solved by the genetic algorithm has not been seen. The paper improves the mathematical model established by Deng Wei[7], designs the customer clustering assignment algorithm and the genetic algorithm, and proves the validity of the algorithm by the experiments.

## 2 Problem description and mathematical model

$m$-VRPTW can be stated as follows. Find a set of closed routes, for a fleet of $|V|$ vehicles ($V$=set of available vehicles from the depot), servicing a set of customers, from each depot at the minimum cost and the least vehicle number when there are enough vehicles to service all the customers; otherwise, find a set of closed routes at the most number of customers. $C$ is the set of the customers. Indices $i$, $j$ refer to customers and depots, while index $i=0\,(or\ i=n+1)$ refers to the depot; an additional index $k$ counts the vehicles ($k$=1, …,$|V|$), the vehicles are initially located at the central depot. Each customer $i$ poses a demand $d_i$, and is bounded by the time window $[ET_i, LT_i]$ that models the earliest and latest time that customer $i$ can be serviced by a vehicle. Furthermore, a service time, $s_i$ is required for each customer $i$. Each route originates and terminates at the depot, while each customer is serviced exactly by one vehicle. There is a travel time $t_{ij}$ and a distance $c_{ij}$ associated with the path from customer (depot) $i$ to customer (depot) $j$. Each route must satisfy the capacity and the time window constraints. The capacity constraint states that the total quantity of goods delivered can not exceed the vehicle capacity $q$. On the other hand, the time window constraint imposes the following condition: a vehicle can

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not service customer $i$ before the lower bound $ET_i$ and after the upper bound $LT_i$. In addition, $N = C \cup \{0, n+1\}$, and $s_{ik}$ is the time that the vehicle $k$ services the customer $i$.

The mathematical programming formulation of the vehicle routing problem with time windows and a limited number of vehicles requires a group of variables, which models the sequence in which vehicles visit customers, and it is defined as follows:

$$x_{ijk} = \begin{cases} 1 & \text{vehicle } k \text{ travels from customer } i \\ & \text{to customer } j \\ 0 & \text{otherwise} \end{cases}$$

Given the above-defined variables and parameters, the problem can be formulated as follows:

$$\min\left[-\sum_{k\in V}\sum_{i\in C}\sum_{j\in N} x_{ijk}, \sum_{k\in V}\sum_{i\in N}\sum_{j\in N} c_{ij}\cdot x_{ijk}, \sum_{k\in V}\sum_{j\in C} x_{0jk}\right] \quad (1)$$

s.t:

$$\sum_{k\in V}\sum_{j\in N} x_{ijk} \le 1 \qquad \forall i \in C \quad (2)$$

$$\sum_{i\in C} d_i \sum_{j\in N} x_{ijk} \le q \qquad \forall k \in V \quad (3)$$

$$\sum_{i\in N} x_{0ik} = \sum_{i\in N, i\ne n+1} x_{i,n+1,k} \le 1 \qquad \forall k \in V \quad (4)$$

$$\sum_{i\in N} x_{ihk} - \sum_{j\in N} x_{hjk} = 0 \qquad \forall h \in C, \forall k \in V \quad (5)$$

$$s_{ik} + t_{ij} - I(1 - x_{ijk}) \le s_{jk}$$
$$\forall i \in N, \forall j \in N, i \ne n+1, j \ne 0, \forall k \in V \quad (6)$$

$$ET_i \le s_{ik} \le LT_i \qquad \forall i \in N, \forall k \in V \quad (7)$$

$$x_{ijk} \in \{0,1\} \qquad \forall i \in N, \forall j \in N, \forall k \in V \quad (8)$$

The first objective function is the most number of customers that are serviced, the second objective is the lowest total distance, and the third objective is the least number of vehicles required.

Constraint (2) states that each customer is serviced once at most; constraint (3) ensures that the accumulated demand on a route can not exceed the vehicle capacity. the next two set constraints ((4) and (5)) state that each vehicle leaves the depot, after arriving at a customer the vehicle leaves again, and finally arrives at the depot; the inequality (6) states that a vehicle $k$ can not arrive at $j$ before $s_{ik}+t_{ij}$ if it is traveling from $i$ to $j$, here $I$ is a large scalar, which makes the constraint trivial to fulfill in case $x_{ijk}=0$; in addition, $t_{ij}$ is the sum of the servicing time $s_i$ and the traveling time from $i$ to $j$; constraint (7) is the time windows constraint.

## 3 Customer clustering assignment algorithm

The function of the customer clustering assignment algorithm is to cluster the customers, to assign the customers to m vehicles, and to construct the vehicle routes to achieve the initialization of genetic algorithm by the route structure heuristic algorithm (the initialization of feasible population is showed in the section 4.4); except for the above function, the customer clustering assignment algorithm in the paper provides a new method for customer clustering assignment of VRPTW. At present, there are many customer clustering assignment algorithms for VRPTW, for example, sweeping algorithm[8], dynamic clustering algorithm[9], priority integrated clustering algorithm[10], and so on, and there are two important parameters needed to be set.

(1) The number of categories, which is the number of vehicles; in $m$-VRPTW, the number of vehicles is given, so it is unnecessary to make the number of categories. There is a problem arisen because of the known number of vehicles, which is that the sum of the capacity of all vehicles may be smaller than the sum of all customers. If all customers are clustered by the above customer clustering assignment algorithms, it is not reasonable that customer clustering assignment's termination condition is the vehicle's capacity, which is that there are many customers who do not belong to any category, and it is difficult to confirm the appropriate vehicle to service them. In the paper, considering the characteristic of $m$-VRPTW, the vehicle capacity that is used as the termination condition is amended. In the literature[11], the formula for determining the number of vehicles is as follows:

$$m = \left[\sum d_i / aq\right] + 1 \qquad i \in C \quad (9)$$

With $m$ being the number of vehicles required; [...] being the integer part, $q$ being the vehicle capacity; $d_i$ being the demand of the customer $i$; $C$ being the customer set; $a$ being the parameter: $0<a<1$, the more the constraints are, the more complex the goods loading(unloading) is and the smaller the value of $a$ is. Based on the formula (9) for determining the number of vehicles required, the vehicle capacity $q_{re}$ as the termination condition is described as follows:

$$q_{re} = \frac{\sum d_i}{a\cdot(m-1)} \quad (10)$$

(2) The distance between the customers. The above clustering assignment algorithms cluster the customers according to the customers' geographic coordinates or the Euclidean distances between customers, and the time window information is ignored. Considering the importance of time window in VRPTW, in the paper, based on the consanguinity that is defined by Hong Lianxi[12], the distance between customers is redefined:

$$dis(i,j) = Aff(i,j)/dis_{ij} \qquad i,j \in C, i \ne j \quad (11)$$

With $Aff(i,j)$ being the appetency between the customer $i$ and the customer $j$. Appetency is relevant with the following factors: (1) $dis_{ij}$, the euclidean distance between the customer $i$ and customer $j$; (2) $TwDis_{ij}$, time window distance between the customer $i$ and customer $j$. The $Aff(i,m)$ can be stated as follows:

$$Aff_{i,j} = e^{-(\varphi Dis_{ij} + \gamma TwDis_{ij})} \qquad i,j \in C, i \ne j \quad (12)$$

With $\varphi$ and $\gamma$ being penalty, used to adjust each factor's magnitude.

Considering the above discussion, based on the classic hierarchical clustering method, a new customer clustering assignment algorithm is designed, and its procedure is as follows:

(1) Every customer is set as a cluster, which is that there are $n$ clusters.

(2) Calculate the distance between the clusters, merge two clusters, whose distance is nearest and the sum of whose demand is less than $q_{re}$, into one cluster.

(3) Calculate the distance between the new cluster and the rest clusters, and then merge two clusters, whose distance is nearest and the sum of whose demand is less than $q_{re}$, into one cluster; and if the number of the clusters is still larger than the vehicle number $m$, repeat the above steps until all the customers are merged into $m$ clusters.

The distance between the clusters is identified by the completely link method, and the distance between the two clusters is the distance $dis(i,j)$ between the farthest customers.

# 4 Genetic algorithm

### 4.1 Chromosome representation

Many chromosome representations are applied to the vehicle routing problem, in which the decimal code based on vehicle routes[13,14,15], and the code based on customers[16,17,18] are intuitive. The codes above are used to solve the vehicle routing problem, in which the number of vehicles is unlimited, and all the customers are serviced, and they can not satisfy the model's requirement, because the number of the vehicle is limited in the model of the paper.

A new chromosome code is developed from the code based on customers designed by Zou Tong[18], to achieve the following two objectives with the model:

If there are enough vehicles that can service all the customers, the objective is to minimize the distance traveled by the vehicles and the number of the vehicles required.

Otherwise, the objective is to maximize the number of the customers that are serviced, and to minimize the distance traveled by the vehicles.

The chromosome representation designed by Zou Tong[18] is stated as follows: the chromosome representation is the sequence of the genes, which is made up of two parts: vehicle number Vehicle-Num, sequence number Sort-Value, denoting that the customer $i$ is serviced by the vehicle Vehicle-Num, and its sequence in the route is determined by the all customers' Sort-Value. In our approach, to solve the situation that there may be no enough vehicles to service all the customers, the value of customer sequence number Sort-Value is restricted: if the customer can be serviced, its value of sequence number Sort-Value is natural number, otherwise, that is 0. Compared with the other chromosome representation, that of our approach does not denote the route of each vehicle intuitively, but all the customers that are serviced by the same vehicle are sorted by Sort-Value, and the route of each vehicle is easily gotten.

### 4.2 Fitness function

The value of the chromosome is evaluated by the fitness assignment based on ranking, and the fitness function [19] is defined as follows:

Step 1: the objective value of the individual is calculated according to the objective function. In our paper, there are three objectives for each individual: $\left[-\sum_{k\in V}\sum_{i\in C}\sum_{j\in N} x_{ijk}, \sum_{k\in V}\sum_{i\in N}\sum_{j\in N} c_{ij} x_{ijk}, \sum_{k\in V}\sum_{j\in C} x_{0jk}\right]$.

Step 2: the individuals are sorted by the first objective $-\sum_{k\in V}\sum_{i\in C}\sum_{j\in N} x_{ijk}$. If some individuals have the same objective value, they are sorted by the second objective $\sum_{k\in V}\sum_{i\in N}\sum_{j\in N} c_{ij}\cdot x_{ijk}$, and the individuals are sorted by the next objectives as the above actions. If some individuals can not be sorted finally, they are sorted randomly.

Step 3: the values of the chromosomes are evaluated by comparing their performance in each objective function through the sorting created by the objective function. The fitness function is defined as follows[20]:

$$E(x_i) = \begin{cases} (N - R(x_i))^2 & R(x_i) > 1 \\ kN^2 & R(x_i) = 1 \end{cases}$$

$N$ is the number of the individuals, $x_i$ is the $i$-th individual in the population, $R(x_i)$ is the sequence number of $i$-th individual after step1 and step3, and $k$ is a constant between 1 and 2, used to increase the fitness when the individual is optimal.

### 4.3 The handling constraint method

In the calculating process, the individual may be infeasible solution because of the violation of the time window and capacity constraints. As the handling constraint method, the punishment strategy is widely used, which is that the penalty is imposed on the infeasible solution to increase the objective function value, and after the constant iterations, the population gradually converges to the optimal feasible solution.

Tests show that when the chromosome code is the above code based on customers and the handling constraint method is the punishment strategy, it is not ideal to deal with the large-scale $m$-VRPTW, which is because the quality of solution relies on the selection of the penalty factor in the punishment strategy, and if the penalty factor is improper, the algorithm may converge at the infeasible solution. The handling constraint method of the paper is that of the literature[21]. This method searches the solution space of the problem though the admixture crossover of feasible and infeasible solutions, and does the selection and operation on feasible and infeasible populations, respectively. It avoids the difficulty of selecting the penalty factor in penalty strategy and makes the handling constrain simplify. Experiment tests show that it is effective to solve

$m$-VRPTW using the code based on customers.

**4.4 Population initialization**

The Vehicle-Num and Sort-Value in the literature[18] is generated randomly when generating the initial population. It is proved by the experiment that this approach for the large-scale $m$-VRPTW convergences slowly, which is because the rate that there are good initial modes is low, if all the Vehicle-Num and Sort-Value are generated randomly. In our approach, considering the customer clustering assignment algorithm, the feasible population is initialized by the route construction algorithm, which is given as follows: firstly, $m$ customer clustering sets $CustSet_i$ are gotten by the customer clustering assignment algorithm, the customers' Vehicle-Num in set $CustSet_i$ are all evaluated as $i$ to accomplish the initialization of Vehicle-Num; for every $CustSet_i$, the customer that satisfies the time window constraint, is chosen randomly from it, and then the customer in the $CustSet_i$ is inserted in turn by the greedy algorithm until the time window constraint or capacity constraint is not satisfied, with the shortest distance in the greedy algorithm being $dis(i, j)$ defined by the formula (11); the customers in $CustSet_i$ are sorted, finally the sort of assignment is evaluated to Sort-Value, and the customers that are not serviced are evaluated as 0, to accomplish the initialization of the feasible population.

For the initialization of the infeasible population, the Vehicle-Num and Sort-Value are randomly generated in the paper.

**4.5 Genetic algorithm manipulation**

(1) Selection

Selection is used to implement the survival of the fittest individual, which is that the individuals in the current population are copied to the new population according to the probability that is proportional to the fitness, and the mating pool is gotten (the middle population between current generation and the next generation). Selection operator is used to increase the average fitness. In the paper, selection operator combines the roulette wheel selection and the elite to retain strategy to accelerate convergence.

(2) Crossover

The algorithm can generate new individuals by crossover operator. Firstly, the individuals in the mating pool are mated randomly; then, the genes in the individuals mated are exchanged each other. From the aspect of generating new individuals, the crossover operator is the primary approach, which determines the global search capability of GA. Crossover operator is related to the problem, and corresponds with chromosome representation. In our approach, crossover operator mixes one-point crossover with two-point crossover.

(3) Mutation

Mutation is to change a gene or a small group of genes by a small probability. From the aspect of generating new individuals, the mutation operator is an assistant method of generating the new individuals, which is indispensable, because it determines the genetic algorithm local search capability. Different from the mutation operation of binary coding, the mutation in our approach is that genes of individuals are randomly generated, which is generating the genes including Vehicle-Num and Sort-Value.

**4.6 Genetic algorithm procedure**

step 0: initialization, set the crossover rate $P_x$, the mutation rate $P_m$, the feasible population size $N_1$, the infeasible population size $N_2$, and the generation span *Gen*; according to the constraints, divide the initial population into the feasible population *popf* and the infeasible population *popinf*.

step 1: operate the crossover and the mutation on the feasible population and the infeasible population.

step 2: according to the constraints, divide the new population after crossover and mutation, into the feasible population *popf* and the infeasible population *popinf*.

step 3: selection operation: according to the fitness of the individual, select $N_1$ individuals which are good enough to form new feasible population from the old population and the new population; so is the new infeasible population that has $N_2$ individuals.

step 4: judge whether the algorithm reaches the generation span *Gen*? NO: go to step 1; YES: terminate procedures.

## 5 Experimental result

According to the above algorithm procedure, the algorithm is implemented using Matlab language. Two experiments are carried: experiment 1 is VRPTW, and experiment 2 is $m$-VRPTW. The results below are based on the following set of GA parameters: the feasible population size $N_1$=30, the infeasible population size $N_2$=30, crossover rate $P_x$=0.8, mutation rate $P_m$=0.3, generation span *Gen*=200.

**Tab.1 The customer data**

| $i$ | $(x_i,y_i)$ | $d_i$ | $[ET_i, LT_i]$ |
|---|---|---|---|
| 0 | (99,39) | 0 | [0,1000] |
| 1 | (39,70) | 0.102 | [74,124] |
| 2 | (22,15) | 0.113 | [58,108] |
| 3 | (97,79) | 0.095 | [15,65] |
| 4 | (6,28) | 0.131 | [96,146] |
| 5 | (62,79) | 0.029 | [47,97] |
| 6 | (42,88) | 0.306 | [85,135] |
| 7 | (86,3) | 0.532 | [21,71] |
| 8 | (87,61) | 0.617 | [9,59] |
| 9 | (88,91) | 0.232 | [37,87] |

Experiment 1 is VRPTW. The customers' data comes from the literature [22], and the details can be seen from tab 1.

The minimum cost of example 1 is 459.18, and the optimal route is 0-3-9-5-6-1-0-8-0-7-2-4-0, which is better than that of the literature [22] (521.59), and is the same with that of the literature [23] (459.18).

Experiment 2 is $m$-VRPTW. C101 and C106[24] are given as Tab.2.

**Tab.2 The optimal solution of the genetic algorithm**

| | Vehicle number | 10 | 9 | 8 | 7 | 6 | 5 | 4 |
|---|---|---|---|---|---|---|---|---|
| C101 | The result of the literature [5] | 100 | 92 | 84 | 75 | 66 | 57 | 47 |
| | The result of the literature [6] | 100 | 92 | 84 | 75 | 66 | 57 | 45 |
| | The result of the genetic algorithm | 100 | 92 | 84 | 75 | 66 | 57 | 46 |
| | Vehicle number | 10 | 9 | 8 | 7 | 6 | 5 | 4 |
| C106 | The result of the literature [5] | 100 | 93 | 86 | 77 | 68 | 58 | 47 |
| | The result of the literature [6] | 100 | 93 | 85 | 76 | 67 | 57 | 47 |
| | The result of the genetic algorithm | 100 | 93 | 84 | 76 | 67 | 56 | 47 |

From the above data, the optimal solution gotten by the algorithm of the paper generally corresponds with that of the literature [6] and [7]; although the optimal solutions of some examples are worse than those of literature [6] and [7], it is only two customers lacked at most, and the reason is as follows: in our approach, the handling constraint method is to divide the population into the feasible population and the infeasible population, and to do genetic algorithm manipulation respectively, although it avoids the difficulty of selecting the penalty factor in penalty strategy and makes the handling constrain simplify, in the implementation of the algorithm, many individuals that have good modes, are partitioned into the infeasible population, because they violate the constraints; when selection is operated without considering the constraint, many individuals' fitnesses in the infeasible population, are bigger than those of the individuals that have the good modes but violate the constraint, so the individuals that have the good modes can't be used in the next generation, resulting in that the good modes are disposed.

## 6 Conclusion

This paper improves the mathematical model for the vehicle routing problem with time windows where a limited number of vehicles is given, which not only reflects the feature of the limited number of vehicles in the depot, but also is compatible with VRPTW; redefines the distance between the customers used in the customer clustering, designs the customer clustering assignment algorithm, which can help achieve the initialization of the genetic algorithm, and provides a new method for VRPTW customer clustering assignment; proposes an improved chromosome representation on the basis of customer, which can denote the different vehicle number sent by the depot and the customer unserviced; designs the genetic algorithm to solve $m$-VRPTW, and proves the validity of the algorithm by the experiments. In addition, vehicle routing problem with time windows and a limited number of vehicles is not only the same with the vehicle number limit, but also is useful to the instance that there are no enough cargo in the depot to service all the customers, and what need to be done, is that the cargo is assigned to the vehicles, which is the transmutation of $m$-VRPTW. Many issues discussed in this paper need to be discussed further, which is our next research, such as to design the better handling constraint method.

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